RPractical_Labordo

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###A1

```
v <- seq(6,55, by = 5) -1
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

###A2

```
w <- rep(c(1,2,3),4)
z <- rep(c(1,2,3), each = 4)
```

###A3

```
a <- 17
b <- 5
c<-2
d<- 144
a%%b
```

```
## [1] 2
```

```
a/b
```

```
## [1] 3.4
```

```
c ^ 10
```

```
## [1] 1024
```

```r
sqrt(d)
```

```
## [1] 12
```

```r
round(pi, 2)
```

```
## [1] 3.14
```

###A4

```r
a <- c(2,4,6,8)
b <-c(1,3,5,7)
a+b
```

```
## [1]  3  7 11 15
```

```r
a*b
```

```
## [1]  2 12 30 56
```

```r
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

###A5

```r
a <- "Student-1"
b <- "Student-2"
c <- "Student-3"
students <- paste(a,b,c)
d <- c(42,42L,"42",41i, TRUE)
class(d)
```

```
## [1] "character"
```

###B1

```r
students <- data.frame(

  id = c(1,2,3,4,5,6,7,8),
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

###B2

```
students <- data.frame(

  id = c(1,2,3,4,5,6,7,8),
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : num  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```r
summary(students)
```

```
##        id              name               sex               section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

```r
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

###B3

```r
students <- data.frame(

  id = c(1,2,3,4,5,6,7,8),
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
students$average <- round((students$midterm + students$final) / 2,1)
students
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

###B4

```r
students <- data.frame(

  id = c(1,2,3,4,5,6,7,8),
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
students$average <- round((students$midterm + students$final) / 2,1)
students$passed <- students$average >= 70 & students$attendance >= 75

students
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5  Elif   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7 Grace   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

###B5

```r
students <- data.frame(

  id = c(1,2,3,4,5,6,7,8),
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
students$average <- round((students$midterm + students$final) / 2,1)
students$passed <- students$average >= 70 & students$attendance >= 75
students$standing <- ifelse(students$average >= 90, "Honors",
                            ifelse(students$average >= 75, "Good", "Needs Support"))
students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

###B1

```
classify <- function(score) {
  if (score >= 85) {
    return("Distinction")
  } else if (score >= 70){
    return("Merit")
  }else if (score >= 50) {
    return ("Pass")
  } else {
    return ("Fail")
  }
}

test_results <- sapply(c(90, 73, 55, 40), classify)
test_results
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

```
students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

###C1

```
new_students <- data.frame (
  id = c(9,10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81,70),
  final = c(84, 68),
  attendance = c(92,80),
  stringsAsFactors = FALSE
)
new_students$average <- round((new_students$midterm + new_students$final) / 2,1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                                ifelse(new_students$average >= 75, "Good", "Needs Support"))
students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
```

```
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

###C2

```
new_students <- data.frame (
  id = c(9,10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81,70),
  final = c(84, 68),
  attendance = c(92,80),
  stringsAsFactors = FALSE
)
new_students$average <- round((new_students$midterm + new_students$final) / 2,1)
new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
                                ifelse(new_students$average >= 75, "Good", "Needs Support"))
students_all <- rbind(students, new_students)
quiz <- c(8,7,10,6,7,9,10,5,8,6)
students_all <- cbind(students_all, quiz)
class(quiz)
```

```
## [1] "numeric"
```

```
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
##    quiz
## 1     8
## 2     7
## 3    10
## 4     6
## 5     7
## 6     9
```

```
## 7    10
## 8     5
## 9     8
## 10    6
```